AI Summit Hackathon 2026

Official Problem Statements: Voice Agents & Automation Track

Welcome to the Voice & Automation track, sponsored by **ElevenLabs** and **n8n**. Your goal is to build fully functional, real-world Minimum Viable Products (MVPs) that combine dynamic voice synthesis, conversational AI, and robust workflow orchestration.

Sector: Workplace Productivity & Engineering

1. Hands-Free Voice Notes to Structured Task Dashboard

PRODUCTIVITY

The Concept: Busy professionals lose track of daily thoughts, meeting action items, and reminders. A user places a web voice call via ElevenLabs and rambles freely (e.g., *"Remind me to email the sponsor deck tomorrow at 2 PM and also buy coffee"*). The system automatically cleans the transcript, extracts tasks/dates, and creates organized entries.

Tech Integration Strategy:

- **ElevenLabs:** Hosts the conversational web widget, captures the raw audio, and streams the user's intent to the backend.
- **n8n:** Receives the webhook from ElevenLabs, formats the extracted tasks using a basic LLM node, and natively connects to the Notion API or Google Calendar API to log the structured data.

2. Voice-Triggered GitHub Bug & Issue Logger

DEVELOPER TOOLS

The Concept: Developers on the move often spot bugs but lack the time to fill out complex tracking tickets. A user calls an AI assistant and speaks a bug description. The AI asks follow-up questions (e.g., *"What browser were you using?"*), and upon verbal confirmation, automatically creates a formatted ticket.

Tech Integration Strategy:

- **ElevenLabs:** Acts as the conversational agent, dynamically asking for missing context before triggering the final tool call.
- **n8n:** Processes the tool-call payload, structures it into markdown, and uses the GitHub or Jira API node to publish the issue to the team's repository.

3. Voice-Activated Daily Standup Bot

TEAM OPS

The Concept: Remote teams suffer from "standup fatigue" where typing out daily updates is tedious. Users press a record widget and speak their update naturally. The bot parses the audio into the classic "Yesterday, Today, Blockers" format and posts it directly to the team's communication channel.

Tech Integration Strategy:

- **ElevenLabs:** Transcribes the spoken update and replies with a friendly confirmation message (e.g., *"Got it, posting to Slack now!"*).
- **n8n:** Intercepts the transcript, restructures it, and uses the Slack or Discord API node to post the summarized bullet points into the correct channel.

Sector: Media, Content & Crisis Management

4. Interactive Voice Podcast Host & Newsletter Reader

MEDIA

The Trend: Audio-first content consumption. Substack posts are static and basic text-to-speech sounds robotic.
The Build: An interactive audio newsletter. The user pastes an article link. The system extracts the content, summarizes it into a conversational script, and renders a realistic, two-person podcast format. Listeners can interrupt the audio via a widget to ask follow-up questions out loud.

Tech Integration Strategy:

- **n8n:** Scrapes the provided URL (using HTTP/HTML extract nodes), routes it through an LLM to generate a dual-host dialogue payload, and fires it to the voice generation pipeline.
- **ElevenLabs:** Uses Conversational AI to render distinct dynamic voices and handles live user voice interruptions/Q&A over the web widget.

5. Automated Emergency Voice Broadcast & Safety Bot

EMERGENCY RESPONSE

The Concept: In campus emergencies, text blasts are often missed. An admin triggers an alert. The system automatically places voice calls (or voice notes) to staff, asks if they are safe, and logs their spoken responses to a live dashboard.

Tech Integration Strategy:

- **ElevenLabs:** Generates the ultra-low latency voice alert and handles the conversational back-and-forth asking for the user's safety status.
- **n8n:** Triggers the mass dispatch via webhooks, aggregates the spoken responses (Safe / Need Assistance), and updates a live Google Sheet or Telegram channel for first responders.

Sector: Deep Research & Financial Intelligence

6. Multi-Agent Financial/Market Research Swarm

FINANCE

The Concept: Deep market research requires cross-referencing news, filings, and trends. This system deploys multiple specialized agents (Data Collector, Analyst, Risk Assessor) to automatically generate a structured investment memo, which is then delivered to the user as an executive audio briefing.

Tech Integration Strategy:

- **n8n:** Acts as the orchestrator. Uses advanced AI Agent nodes to trigger web searches (Tavily/Serper), pull financial APIs (Yahoo Finance), loop through multiple agent reviews, and compile the final text memo.
- **ElevenLabs:** Takes the final synthesized memo and uses the Text-to-Speech API to generate a highly professional, studio-quality audio briefing that the user can download.

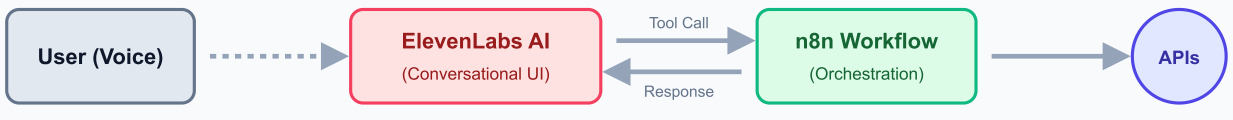
7. Deep Research & Automated Literature Review Agent

RESEARCH

The Concept: Researchers take days to aggregate academic papers. This agent takes a complex prompt, recursively searches web sources, extracts findings, generates a cited research report, and produces an interactive audio-explainer where the user can debate the findings.

Tech Integration Strategy:

- **n8n:** Manages the Retrieval-Augmented Generation (RAG) pipeline. Pulls PDFs, chunks data into a vector database node, and formats the literature review.
- **ElevenLabs:** Functions as the "Research Assistant Avatar," using the Conversational API to let the user verbally question the research findings, triggering n8n tool-calls to check citations in real-time.



Standard Hackathon Reference Architecture